CLOUD COMPUTING TA-2

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Section: A2

Roll no : 27

Event-Driven Serverless File Processing using AWS CloudFormation

Execution Report and Deployment Evidence

Cloud platform	**Amazon Web Services (AWS)**
Deployment region	Asia Pacific (Sydney), ap-southeast-2
CloudFormation stack	TA2-Serverless-File-Processing
Lambda function	TA2-File-Processor
SNS topic	FileProcessingAlerts
Implementation and testing	26 October 2026

Submission deliverables: template.yaml, architecture-diagram.png, and TA2-submission-report (Word format).

This report records the AWS infrastructure deployed and the successful file-processing test demonstrated using actual AWS screenshots.

Objective

Design and deploy an event-driven serverless file-processing architecture using AWS CloudFormation. When a file is uploaded into an Amazon S3 bucket, an S3 ObjectCreated event invokes an AWS Lambda function connected to a private subnet. Lambda reads the file details, saves processed metadata to S3 and publishes a completion summary through Amazon SNS to a confirmed email subscriber.

AWS Services and Configuration

Amazon S3	**One private bucket with SSE-S3 (AES256), Block Public Access, and s3:ObjectCreated:* event notifications filtered to uploads/.**

Amazon VPC	VPC 10.0.0.0/16; public subnet 10.0.1.0/24; private subnet 10.0.2.0/24; DNS settings enabled.
Network connectivity	Internet Gateway, public NAT Gateway with Elastic IP, and separate public/private route tables.
AWS Lambda	Python 3.13; 128 MB RAM; 60-second timeout; attached to the private subnet.
AWS IAM	Lambda VPC execution policy; S3 GetObject/PutObject permissions; SNS Publish; S3 invocation permission.
Amazon SNS	FileProcessingAlerts topic with a confirmed email subscription.

File Processing Workflow

1. The user uploads uploads/sample.txt to Amazon S3.
2. The s3:ObjectCreated:* event notification invokes the Lambda function.
3. Lambda reads object details and writes processed/sample.txt.json back into S3.
4. Lambda publishes a summary to the SNS topic, which sends an email notification.

Architecture Diagram

The architecture below shows the S3 event-driven processing flow, Lambda in the private subnet, outbound routing through the NAT Gateway and Internet Gateway, and SNS email delivery.

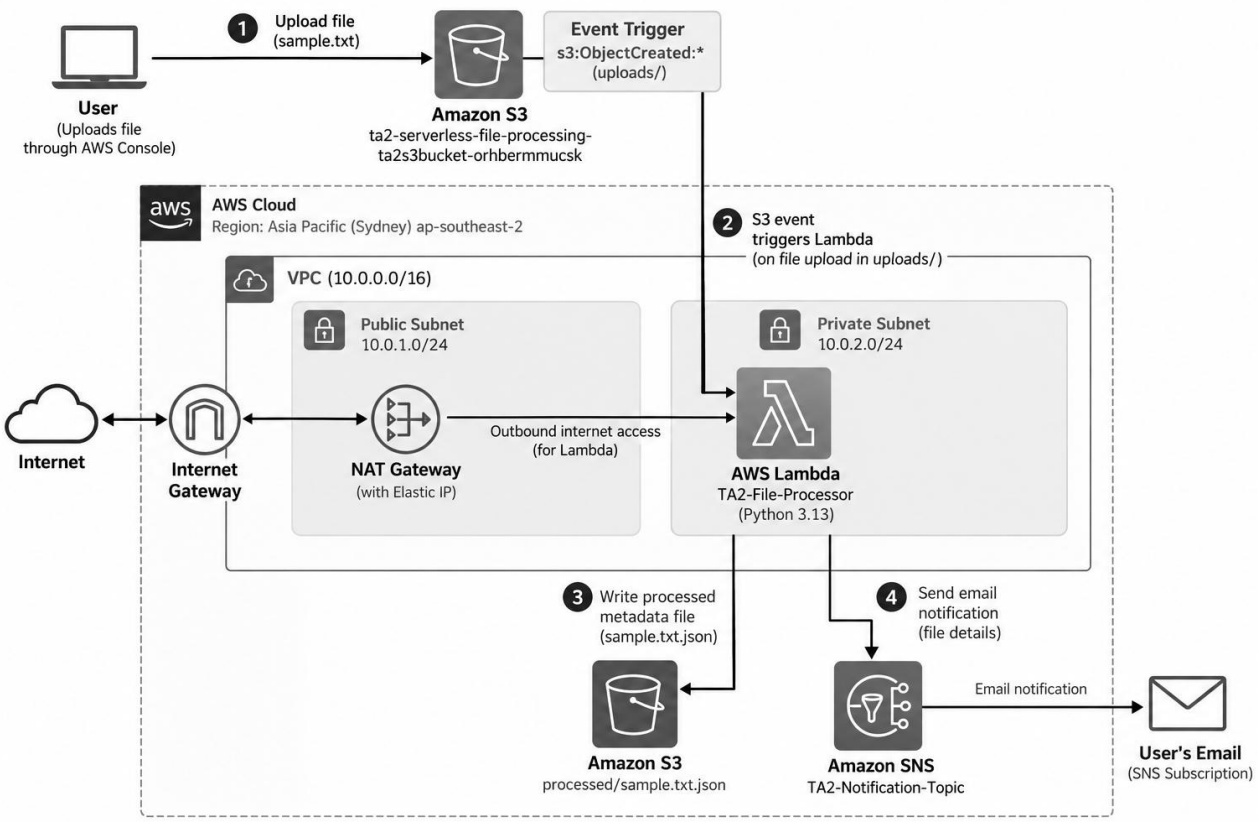


Figure 1. Event-driven serverless file-processing architecture

Implementation Details

Infrastructure as Code

The CloudFormation YAML template provisions the VPC, public and private subnets, Internet Gateway, NAT Gateway, route tables, S3 bucket, SNS topic and subscription, Lambda function, security group, IAM execution role, and resource-based Lambda invoke permission.

Network Configuration

Public subnet	10.0.1.0/24; default route to Internet Gateway
Private subnet	10.0.2.0/24; default route to NAT Gateway
Lambda network placement	TA2PrivateSubnet with TA2LambdaSecurityGroup
Service connectivity	Outbound access through the NAT Gateway

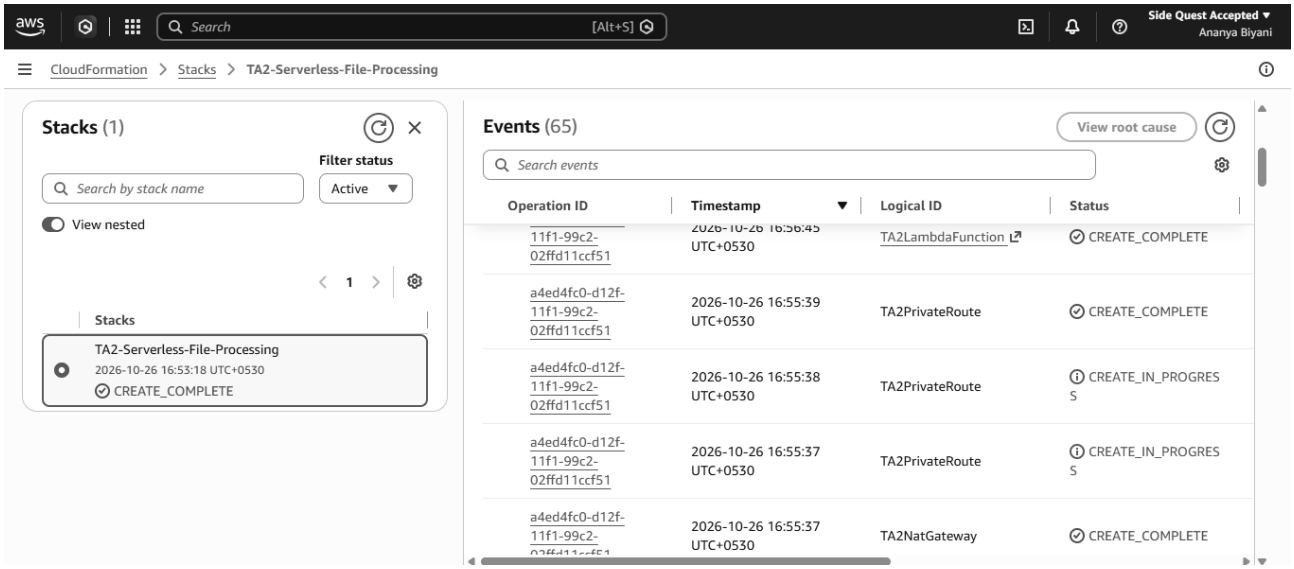
Lambda Processing

The Python handler extracts the S3 bucket and object key from the event, reads the file details, writes a JSON record containing the file name, byte count, processing time and status, then publishes an SNS message. Input objects are stored under uploads/ and output metadata under processed/.

CloudFormation Deployment

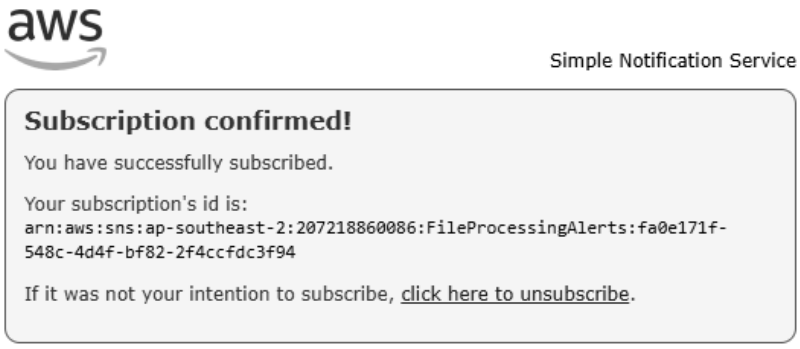
The initial stack reached CREATE_COMPLETE. To configure the S3 event notification safely, the missing Lambda invoke permission was deployed through a separate stack update. The event notification was then added through CloudFormation, and the stack reached UPDATE_COMPLETE.

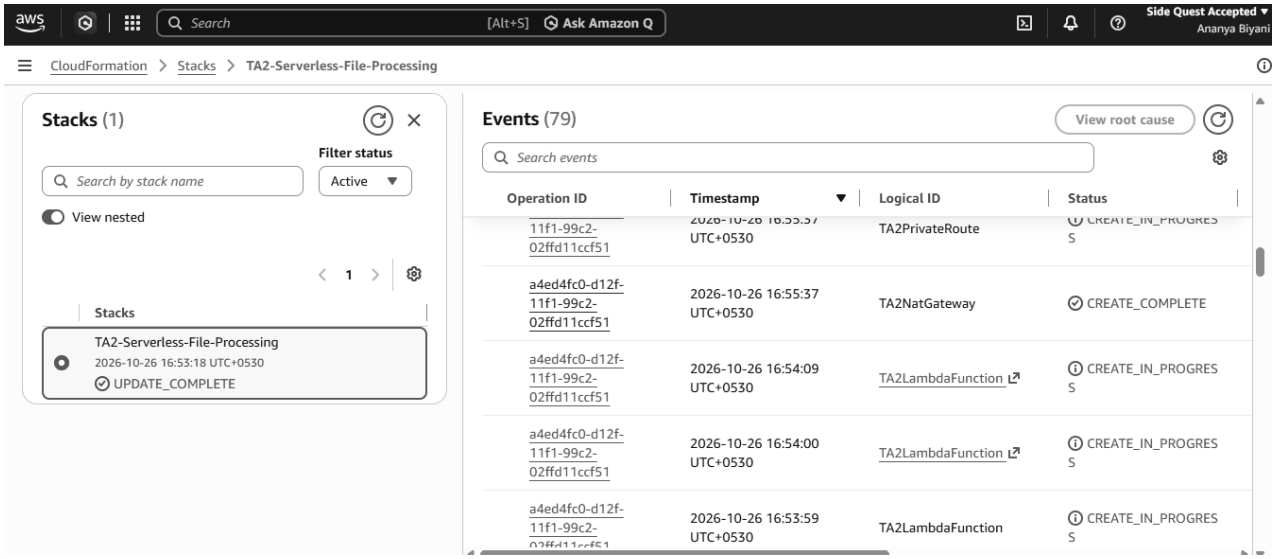
CloudFormation Stack Creation Evidence



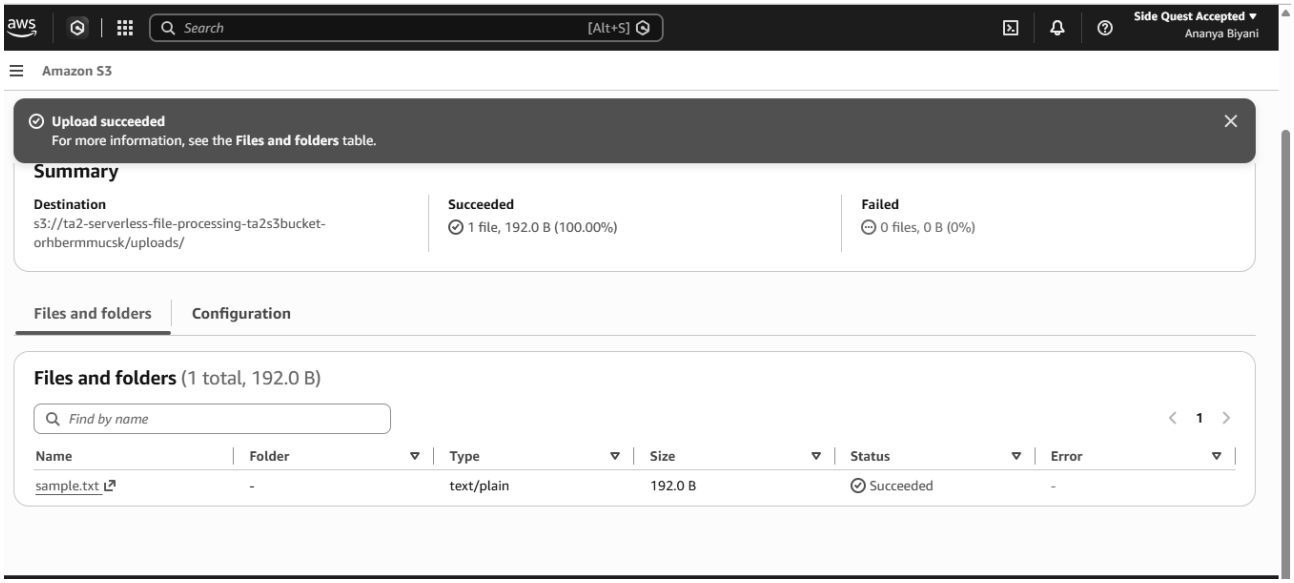
Result: The initial CloudFormation deployment completed successfully in ap-southeast-2.

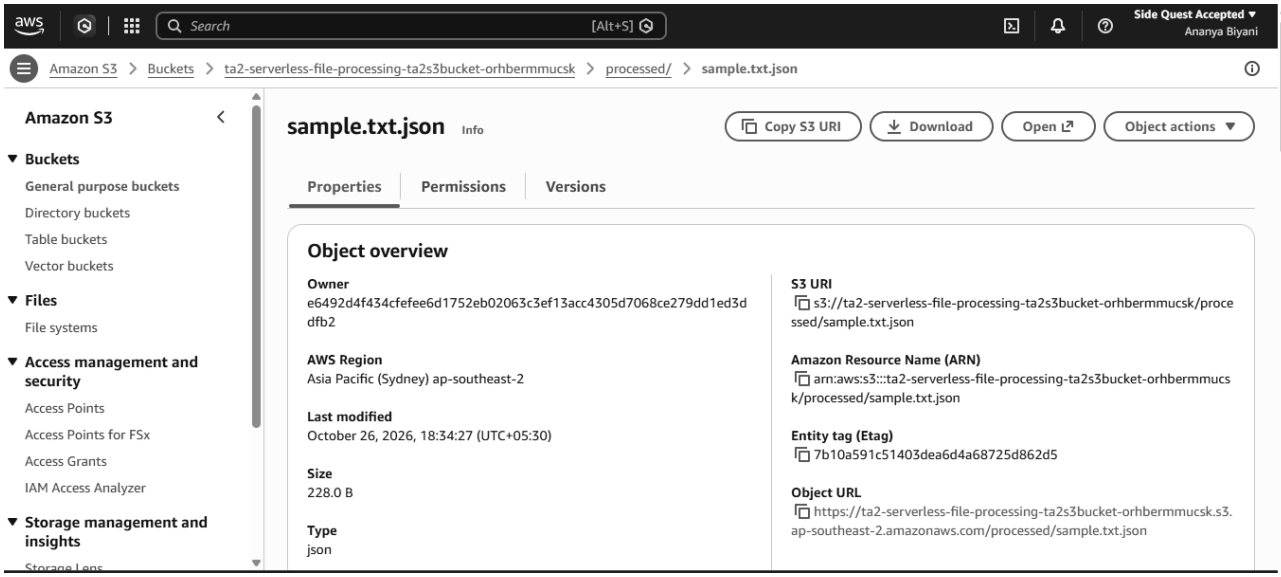
SNS Subscription and Stack Update Evidence



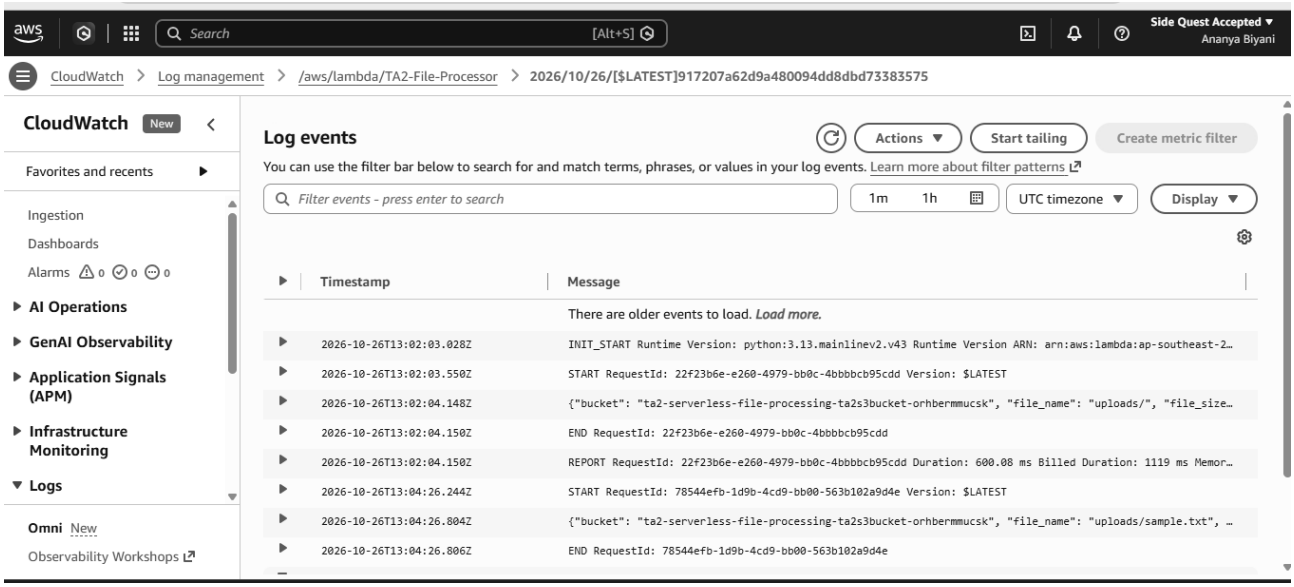


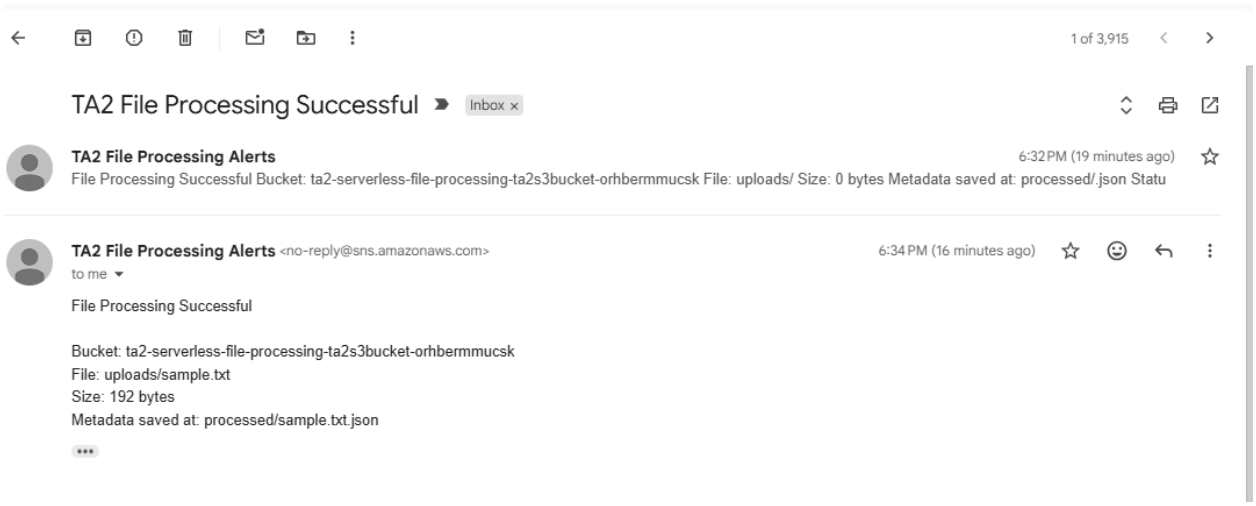
S3 Upload and Processing Evidence





Lambda Execution and SNS Notification





Conclusion

The TA-2 event-driven serverless file-processing system was deployed successfully using AWS CloudFormation. Uploading sample.txt to Amazon S3 invoked the Python Lambda function within the private subnet. The function saved the processed JSON metadata in S3 and sent a successful processing alert through Amazon SNS. CloudFormation status, S3 output, Lambda logs and the received email provide evidence that the required workflow operated successfully.